

Converting Data Types

You cannot always combine variables and constants of different data types. You may have to convert them before an operation can be executed on them.

Automatically Converting Data Types

Plant Simulation automatically converts numeric values. You have to explicitly start all other type conversions.

Plant Simulation automatically converts the data type *integer* to *real* and vice versa. When you assign a *real* value to an *integer* variable, the digits after the decimal point will be lost. When parameters are passed, *Plant Simulation* automatically converts the data type *real* to *integer* and vice versa. The same is true for the data types *dateTime* and *date*. Here, *Plant Simulation* deletes the time part.

During calculations *Plant Simulation* treats the units of the physical data types *length*, *weight*, *speed*, *acceleration* and *time* as *real* numbers. If a calculation results in another physical data type, *Plant Simulation* assigns the result the corresponding data type. The quotient of *length* and *time* is assigned the data type *speed*. If the result cannot be assigned a physical data type, *Plant Simulation* uses *real* instead.

You can only assign a value with the correct unit or a value without a unit (*real* or *integer*) to a *local variable* of physical data types (*length*, *weight*, *speed*, *acceleration*, *time*). In all other cases the assignment throws an error to point out a potential problem.

Note

Automatic conversion from *real* to *integer* will truncate decimal places, similar to the TRUNC function in Excel.

Currently, we provide two workaround for this:

Use the **round** function:

```
var realVal := pi
var intVal : integer := round(realVal)
```

Add the value 0.5:

```
var realVal := pi
var intVal : integer := realVal + 0.5
```